

Inventory Procurement Sales Business Intelligence Platform Executive Summary

Effective inventory and sales management are critical for maximizing profitability in the retail and wholesale beverage industry. This project analyzes inventory, procurement, sales, vendor performance, and profitability using Python, MySQL, and Power BI.

The objective was to identify underperforming products, optimize inventory levels, evaluate vendor contributions, analyze purchasing efficiency, and improve overall profitability.

The analysis was conducted on six datasets containing inventory, purchases, sales, vendor pricing, and invoice information.

Business Problem

The company faces challenges in managing inventory efficiently while maintaining profitability.

The analysis aims to:

- Identify underperforming brands requiring promotional or pricing adjustments.
- Determine top vendors contributing to sales and gross profit.
- Analyze the impact of bulk purchasing on unit costs.
- Assess inventory turnover to reduce holding costs.
- Investigate profitability differences between vendors and products.
- Identify dead stock and overstock inventory.

Dataset Overview

Begin Inventory

Contains opening inventory levels at the beginning of the year.

Features:

- InventoryId
- Store
- City
- Brand
- Description
- Size
- OnHand
- Price

- StartDate
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End Inventory

Contains inventory levels at the end of the year.

Features:

- InventoryId
 - Store
 - City
 - Brand
 - Description
 - Size
 - OnHand
 - Price
 - EndDate
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Purchase Prices

Contains product procurement pricing and vendor information.

Features:

- Brand
 - Description
 - PurchasePrice
 - VendorNumber
 - VendorName
 - Classification
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Purchases

Contains purchase transactions from vendors.

Features:

- VendorNumber
- VendorName
- Quantity

- PurchasePrice
 - Dollars
 - Purchase Dates
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Sales

Contains product sales transactions.

Features:

- SalesQuantity
 - SalesDollars
 - SalesPrice
 - SalesDate
 - Vendor Information
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Vendor Invoice

Contains vendor invoices and freight costs.

Features:

- Freight
 - InvoiceDate
 - PayDate
 - Quantity
 - Dollars
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Technology Stack

Python

Used for:

- Data Cleaning
- Missing Value Handling
- Data Validation
- Exploratory Data Analysis
- Feature Engineering

Libraries:

- Pandas
 - NumPy
 - Matplotlib
 - Seaborn
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MySQL

Used for:

- Data Storage
 - Data Modeling
 - SQL Analytics
 - Aggregations
 - Summary Tables
 - Business Query Analysis
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Power BI

Used for:

- Dashboard Development
 - KPI Tracking
 - Data Visualization
 - Executive Reporting
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Data Cleaning & Preparation

Several preprocessing steps were performed:

- Removed duplicate records.
 - Corrected date formats.
 - Standardized column names.
 - Fixed missing city values in End Inventory using Begin Inventory mapping.
 - Validated inventory and sales relationships.
 - Checked for null values and inconsistencies.
 - Created derived metrics for reporting.
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Exploratory Data Analysis

Inventory Analysis

Key Findings

Total Opening Inventory Value:

\$68.05 Million

Total Closing Inventory Value:

\$79.70 Million

Total Inventory Units:

4.22 Million Units

Inventory increased significantly during the year, indicating additional stock accumulation.

Highest Inventory Value Products

- Jack Daniels No 7 Black
- Grey Goose Vodka
- Ketel One Vodka
- Baileys Irish Cream
- Jameson Irish Whiskey
- Absolut 80 Proof

These products accounted for a large portion of total inventory investment.

Dead Stock Products

Products with inventory but minimal or no sales:

- Bacardi Pineapple Fusion
- Bacardi Coconut

These products represent potential inventory carrying costs and markdown opportunities.

Sales Analysis

Objectives

- Understand revenue trends.
- Identify top-selling products.
- Evaluate store performance.

- Analyze customer demand patterns.
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Key Findings

Revenue Concentration

A relatively small group of products generated a large share of total sales revenue.

Premium liquor brands consistently contributed the highest revenue.

Store Performance

Certain stores significantly outperformed others in revenue generation.

Top-performing stores accounted for a substantial portion of overall sales.

Product Demand

Products with strong inventory turnover demonstrated consistent demand throughout the year.

Vendor Analysis

Objectives

- Evaluate supplier contribution.
 - Assess procurement efficiency.
 - Analyze purchasing behavior.
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Key Findings

Vendor Contribution

Procurement spending was concentrated among a limited number of vendors.

Several suppliers dominated both inventory procurement and sales contribution.

Bulk Purchasing Impact

Analysis indicated that larger purchase quantities generally resulted in lower unit procurement costs.

Bulk purchasing provided cost advantages for high-volume products.

Freight Costs

Freight expenses varied significantly across vendors.

Certain vendors generated higher logistics costs that could impact profitability.

Profitability Analysis

Objectives

- Measure gross profit.
 - Evaluate profit margins.
 - Identify high and low performing products.
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Key Findings

High-Profit Products

Several premium brands generated both high revenue and strong margins.

These products represent core business drivers.

Low-Profit Products

Some products generated reasonable sales volume but relatively low margins.

These products may require:

- Pricing adjustments
 - Vendor renegotiation
 - Promotional optimization
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Vendor Profitability

Vendor profitability varied considerably.

A small group of vendors contributed disproportionately to gross profit.

Dashboard Development

Five interactive Power BI dashboards were developed.

Executive Dashboard

Provides overall business performance.

KPIs:

- Revenue
 - Profit
 - Inventory Value
 - Vendor Count
 - Product Count
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Sales Dashboard

Provides insights into:

- Revenue trends
 - Product performance
 - Store performance
 - Sales seasonality
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Inventory Dashboard

Provides insights into:

- Inventory valuation
 - Inventory turnover
 - Overstock products
 - Dead stock analysis
 - Geographic inventory distribution
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Vendor Dashboard

Provides insights into:

- Procurement spend
 - Vendor dependency
 - Freight costs
 - Payment cycle analysis
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Profitability Dashboard

Provides insights into:

- Gross profit

- Profit margin
 - Product profitability
 - Vendor profitability
 - Store profitability
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Business Recommendations

Inventory Optimization

- Reduce inventory levels for slow-moving products.
 - Liquidate dead stock through promotions and discounts.
 - Improve inventory turnover monitoring.
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Vendor Management

- Reduce dependency on a small group of suppliers.
 - Negotiate improved procurement pricing.
 - Monitor freight costs regularly.
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Product Strategy

- Focus on high-margin products.
 - Reevaluate low-margin product pricing.
 - Promote products with high profitability potential.
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Operational Efficiency

- Improve demand forecasting.
 - Align purchasing decisions with sales trends.
 - Optimize stock allocation across stores.
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Conclusion

The analysis successfully identified opportunities to improve inventory efficiency, optimize procurement spending, and increase profitability.

The combination of Python, MySQL, and Power BI enabled end-to-end analytics, transforming raw transactional data into actionable business insights.

The project demonstrates practical applications of data cleaning, SQL analytics, business intelligence, and dashboard development in a real-world retail inventory environment.